

IND713/MEC713 – FALL 2023

Risk Management Lab

General Instructions

Instructions:

1. Assemble with your team members.
Note: We studied Risk Management in **Week 5**.
2. The standard cover page must be submitted with the lab report.
 - a. No attendance by an individual, no grade for that individual
 - b. No signature or student ID of an individual, no grade for that individual
 - c. No cover page by group, no grade for that group
3. Work through the different parts of the exercise.
4. Lab deliverables must be uploaded to D2L at the end of the Lab with the cover page in front. The name of the section in D2L is “**Lab 6**” or “**Practicum 6**”.
5. The submitted Lab report will be graded based on general appearance, proper grammar and language, thoroughness and quality of answers.

Expected Deliverables

1. Cover page
2. List if identified risk and description of risks
3. Probability of occurrences
4. Impacts
5. Risk matrix
6. Critical risks
7. Responses to critical risks

Suppose that you are the project manager of a building complex project, and you want to perform a risk management process to identify, analyze, and respond the project risks. The project is located in downtown Toronto, something like TRS building (Ted Rogers School of Management).

Step 1) Risk Identification. With your team, try to identify at least 10 risks. Remember, Risk is an **uncertain** event that may have a negative (or positive) effect on project objectives (Time, Cost, Quality, and Scope). List the risks and describe them in the following Risk Register Form, which will be completed through the next steps.

Risk Register Form

ID	Risk	Risk Description	Probability of occurrence	Impact			Risk Level
				Schedule	Cost	Quality	

Step 2) Qualitative Risk Analysis. For each risk, estimate the probability of occurrence and impact (on project time, cost, and quality), using the following scales.

Risk Probability Definitions

Probability	Definition
<i>Frequent</i>	Risk event expected to occur
<i>Likely</i>	Risk event more likely than not to occur
<i>Occasional</i>	Risk event may or may not occur
<i>Seldom</i>	Risk event less likely than not to occur
<i>Improbable</i>	Risk event not expected to occur

Risk Impact Definitions

Project Objective	Very Low	Low	Moderate	High	Very High
Cost	Insignificant cost impact	< 10% cost impact	10-20% cost impact	20-40% cost impact	> 40% cost impact
Schedule	Insignificant schedule impact	< 5% schedule impact	5-10% schedule impact	10-20% schedule impact	> 20% schedule impact
Quality	Barely noticeable	Only very demanding applications impacted	Sponsor must approve quality reduction	Quality reduction unacceptable to sponsor	Product becomes effectively useless

Using the probability and maximum impact on objectives, put the risks in the following Probability-Impact Matrix so that you can prioritize them in three levels: Minor, Moderate, and Major.

Risk Probability and Impact Matrix

Probability	Impact				
<i>Frequent</i>					
<i>Likely</i>			Moderate	Major	
<i>Occasional</i>	Minor				
<i>Seldom</i>					
<i>Improbable</i>					
	Very Low	Low	Moderate	High	Very High

Step 3) Risk Response. Try to find an appropriate response to **Major** risks.

ID	Risk	Response
...		